

# EDUKE

## An Academic Performance Prediction System with AI Chatbot

### Introduction

**Eduke** is an advanced, **AI-powered academic performance prediction system** designed to enhance the learning experience for students, streamline communication among teachers, parents, and students, and provide actionable insights for improving academic performance. The system leverages modern technologies to simplify academic tasks, such as attendance tracking, grade management, performance evaluation, and quiz administration, while also incorporating an AI chatbot for personalized assistance and academic guidance.

The system supports four distinct user roles: **students**, **parents**, **teachers**, and **admin**, each with unique responsibilities. Students can access their grades, attendance records, and evaluations, they can access study tips from the AI chatbot. The AI chatbot analyses their data and tells the student about his/her current academic situation whether he/she should adjust their studying method to make good performance. Students can participate in quizzes arranged by teachers, and their responses are stored and evaluated, providing immediate feedback. Parents can monitor their child's progress, provide evaluations on factors like study time and stress management, and communicate with teachers. Teachers can manage attendance, input grades, evaluate student performance, and conduct quizzes using the system. Admins oversee user account creation, assign classes and subjects, and manage the overall system.

Eduke also incorporates **AI-driven performance prediction**, powered by analysing a combination of academic and behavioral factors that influence student outcomes. Attendance plays a vital role, as consistent participation ensures regular exposure to lessons. Marks from previous assessments and internal exams highlight academic trends and areas of strength or weakness. Parent feedback, including factors like study time, sleep patterns, and stress management, offers a view of the student's home environment, while teacher feedback, covering aspects such as homework completion, class participation, focus, and test preparation, provides classroom-specific insights. Additionally, quiz results and other evaluation metrics, such as ratings for class difficulty and overall performance, are incorporated to create a comprehensive profile. By integrating these diverse data points, Eduke predicts outcomes and offers actionable recommendations to improve academic performance.

### Features

#### For Students

- Participate in quizzes created by teachers and answer multiple-choice questions.
- View detailed quiz results, including correct answers and performance insights.
- Receive personalized assistance from the AI-powered chatbot about their progress, including areas to improve based on quiz results, attendance, and marks.
- Access attendance records, marks, quiz results, and evaluations.
- Interact with teachers through the integrated chat feature for academic discussions or clarifications.

### For Parents

- Access their child's attendance, marks, and evaluations.
- Provide performance evaluation data for factors like study time, sleep patterns, and stress management.
- View their child's quiz performance and insights, helping them support their child's learning effectively.
- Submit an overall "Parent Rating" for their child as part of the evaluation process.
- Communicate with teachers via chat to discuss academic progress and address concerns.

### For Teachers

- Create quizzes with multiple-choice questions, including options and the correct answer.
- Publish quiz results for students to view, including performance summaries.
- Manage attendance and marks records efficiently for their classes.
- Evaluate students on various metrics like focus, participation, homework completion, and test preparation.
- Provide "Teacher Ratings" and input detailed feedback to contribute to evaluations.
- Interact with students and parents through the chat feature for academic discussions and feedback.

### For Admin

- Manage user accounts for students, parents, and teachers, ensuring proper role assignments.
- Assign classes, subjects to teachers and students.
- Oversee the system's operations and generate detailed reports on academic performance, quiz participation, and overall engagement.

### AI-Powered Chatbot

- Assists students with insights into their progress, providing feedback on marks, attendance, quiz performance, and areas of improvement.
- Offers personalized academic advice based on quiz results and evaluations.
- Helps students and parents with study tips and strategies for improving academic outcomes.
- Provides tailored responses based on the user role (student, parent, or teacher), ensuring relevant and meaningful interactions.

## Database Design

Eduke's database is structured with 10 relational tables to store and manage data efficiently:

1. **Institution Table:** Stores details about institution.
2. **Users Table:** Stores common details for all user roles.
3. **Students Table:** Manages student-specific data, including roll numbers and class associations.
4. **Parents Table:** Links parents to their child and stores login credentials.